
HEBCPF

Holomorphic-Embedding-Based Continuation Power-Flow All-Solutions Solver

*Finding all real-valued solutions of the AC power-flow equations
via the ellipsoidal formulation, holomorphic embedding, Padé approximation, and
numerical continuation*

Release: MEX v2.202607

Windows 64-bit (compiled MEX) build

Method: Holomorphic Embedding Based Continuation (HEBC)
Originating work: D. Wu and B. Wang, *IEEE Access*, 2019
Foundations: Elliptical branch tracing — B. Lesieutre & D. Wu,
Allerton 2015;
Algebraic set preserving mappings — D. Wu,
Ph.D. dissertation, UW–Madison, 2017
Document: User Guide and Reference Manual

When you publish results obtained with this solver, please cite the HEBC paper (see Section 13).

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1 Overview

HEBCPF computes **all** real-valued solutions of the AC power-flow (load-flow) equations for a given network, rather than the single operating point returned by a conventional solver. Knowing the complete solution set is essential for transient-stability energy-function methods (locating type-1 unstable equilibria), voltage-stability analysis, and the study of multiple high-voltage operating points.

The solver implements the **Holomorphic Embedding Based Continuation (HEBC)** method of Wu and Wang (see Section 13). The power-flow equations are first recast in an *elliptical form* (Lesieutre and Wu, 2015) so that every solution-tracing curve is a bounded loop; the loops are then followed by a hybrid numerical scheme:

- **Holomorphic embedding** traverses the smooth (non-singular) portions of a curve in a few very large steps, using power-series / Padé evaluation of the holomorphically-embedded voltages;
- a classical **predictor–corrector continuation** takes over only in the neighbourhood of singular points (turning points / folds), where it steps carefully through the fold and hands control back to the holomorphic routine.

Each time the tracing (homotopy) parameter crosses zero, a power-flow solution has been found. Starting from one known operating point (from MATPOWER’s `runpf`), the solver recursively traces the curves emanating from every discovered solution until no new solution appears.

This release adds a **consensus Padé-pole estimator** to the classic method (Section 5.3), which stabilises the placement of the holomorphic/continuation hand-off by agreeing a singularity estimate across several buses rather than trusting a single bus.

Three execution frameworks share the identical numerical hot path:

Framework	Entry script	Best for
Serial	<code>run_batch.m</code>	small cases, debugging, reproducibility
Parallel (parfor)	<code>run_batch_par.m</code>	medium cases, one worker per equation
Parallel (parfeval)	<code>run_batch_parfeval.m</code>	large cases (recommended)

2 Requirements and dependencies

MATLAB (required).

Developed and tested on Windows 64-bit, MATLAB R2022a and later. Uses `sparse`, `linsolve`, `lu`, and standard linear algebra.

MATPOWER (required).

MATPOWER must be installed and on the MATLAB path. The setup stage calls MATPOWER functions including `runpf` for the initial operating point, `makeYbus` for the bus-admittance matrix, and `idx_bus/idx_brch` for MATPOWER matrix indices. The solver reads the bundled MATPOWER-format `caseXXX.m` data files, but does *not* ship `runpf`, `makeYbus`, `idx_bus`, `idx_brch`, or their dependencies — those come from MATPOWER (<https://matpower.org>).

Parallel Computing Toolbox (optional).

Required *only* for the two parallel frameworks (`run_batch_par.m`, `run_batch_parfeval.m`). The serial framework runs without it.

Compiled MEX (shipped).

Two hot-path functions are provided as Windows 64-bit binaries, `Pade_Apprxmt_mex.mexw64` and `holomorphic_cont_tri_mex.mexw64`. These `.mexw64` files are only expected to run on Windows 64-bit MATLAB. They make the serial trace roughly $2\times$ faster than the pure-MATLAB build with identical results. On Linux/macOS, or after editing hot-path source, you must re-build them (Section 10) with MATLAB Coder and a configured C compiler.

3 Installation

1. Install MATLAB and MATPOWER; confirm `runpf` works (which `runpf` returns a path).
2. Unpack this folder (HEBCPF_MEX_v2.202607) anywhere on disk.
3. Add the folder to the MATLAB path each session:

```
>> cd '<path>\HEBCPF_MEX_v2.202607'
>> addpath(pwd)
```

4. Add only this one HEBCPF release folder to the MATLAB path. Do not add all four release folders simultaneously, because they contain many functions and cases with the same names.
5. (Windows x64 only) nothing else is needed — the shipped `.mexw64` binaries are picked up automatically. On other platforms, run `build_mex` (Section 10) or switch to a pure-MATLAB build.

4 Quick start

Before running any driver, confirm MATLAB can find MATPOWER:

```
>> which runpf
>> which makeYbus
```

Both commands should return paths inside the MATPOWER installation.

From a system shell (non-interactive `-batch`):

```
:: SERIAL
matlab -sd "<path>\HEBCPF_MEX_v2.202607" -batch "addpath(pwd); run('run_batch.m')"
```

```
:: PARALLEL (parfor)
matlab -sd "<path>\HEBCPF_MEX_v2.202607" -batch "addpath(pwd); run('run_batch_par.m
    ')"
```

```
:: PARALLEL (parfeval, recommended for big cases)
matlab -sd "<path>\HEBCPF_MEX_v2.202607" -batch "addpath(pwd); run('
    run_batch_parfeval.m')"
```

From an interactive MATLAB session:

```
>> addpath(pwd)
>> run_batch           % serial batch
>> run_batch_par       % parallel (parfor)
>> run_batch_parfeval  % parallel (parfeval)
>> r = run_merged_case('case14mod','parfeval'); % one case, any mode
>> r.solutions, r.max_residual, r.wall_time
```

The two parallel drivers start a local **parpool** automatically (default worker count) if none is running. The one-time pool start-up is excluded from the reported timing.

5 User options

5.1 Choosing which cases to run

Each driver has a list near the top:

```
cases_to_run = { 'case9', 'case14mod', 'case39', ... };
```

Comment out or add case names. Bundled test systems (expected solution counts, collapsing antipodal pairs, in parentheses):

case3 (6)	case4gs (6)	case4BB0 (14)	case4BBc (12)
case5loop (10)	case5Salam (10)	case6ww (6)	case7Salam (4)
case9 (8)	case9Q (8)	case14mod (30)	case33bw (16)
case39 (176)	case30/case_ieee30 (472)	case57mod (606)	case57 (1322)

5.2 Load scaling

Each driver contains

```
factor = 1 + 0.19*(0); % default = 1 (nominal load)
```

Changing the multiplier (the “(0)”) scales all bus loads and generator outputs, letting you sweep load level. Leave it at 1 to reproduce the nominal solution counts.

5.3 Padé pole estimation mode

The hand-off from the holomorphic routine to the continuation routine is driven by an estimate of the nearest singularity, taken from the real roots (poles) of the Padé denominator. This release estimates that pole by **consensus** across a set of buses: the true singularity sits at nearly the same magnitude on every bus, whereas spurious (Froissart) poles are bus-specific. The estimator takes the smallest pole magnitude agreed on by at least half of the sampled buses (clustered within `pole_cl_tol`), falling back to the global smallest if no consensus forms.

Two options select the bus set (in `hybrid_traceloops_4_with_mex.m`, or overridden serially via the global `POLEMODE`):

'fixed' (option 1)

use a predefined bus set `param.pole_fixed_buses` (shipped default 1:2, auto-filtered to $\leq$ bus count).

'random' (option 2, shipped default)

draw `param.pole_nbus` buses (default 3) at random *once per curve* and hold them fixed for that whole trace. Because this folder seeds the random generator deterministically per (`solution`, `equation`) pair (from a solution signature, bus count and equation index), the random set is fully reproducible and independent of parallel worker scheduling.

To select the mode from a serial session:

```
>> global POLEMODE; POLEMODE = 'fixed'; % or 'random'
>> run_batch
```

Parallel workers do not inherit client globals; for the parallel drivers set the default directly on line 48 of `hybrid_traceloops_4_with_mex.m`.

5.4 Programmatic single-case runs: `run_merged_case`

While the `run_batch*` drivers loop over a list of cases and write CSV summaries, `run_merged_case` is the **unified single-case harness**: it runs *one* case in *any* of the three frameworks behind a single function call and returns a structured, machine-readable result instead of printing to the console.

```
[result, solutions] = run_merged_case(case_name, mode, close_pool_on_finish)
% mode = 'serial' | 'parfor' | 'parfeval'
```

It performs the identical preprocessing as the batch drivers (it reuses `main.m` as the setup template, substituting the requested case for the active `mpc=case...` line), traces the case in the selected framework, and **self-verifies** by recomputing the maximum residual $|z^\top M_i z - b_i|$ over every solution z and every equation i . The call never throws: on failure it returns a `FAIL` result rather than erroring. The returned `result` struct contains:

Field	Meaning
<code>status</code>	'PASS' or 'FAIL'
<code>case_name, mode</code>	echoed inputs
<code>buses</code>	number of buses
<code>solutions</code>	number of solutions found
<code>max_residual</code>	largest constraint residual over the solution set (correctness check)
<code>wall_time</code>	trace wall-clock time (excludes pool start-up)
<code>error_id, message</code>	populated only on FAIL

The second output, `solutions`, is the full solution matrix `Zsave` (each column a solution). Typical uses: A/B comparison of the three frameworks on one case (identical setup, compare `wall_time/solutions/max_residual`); automated regression or validation checks; and obtaining the solution set and residual directly as return values. Because it reads `main.m`, that file must remain in the folder for this harness to work.

6 How the solver works

6.1 Elliptical formulation

In rectangular voltage coordinates $x = [V_d^\top V_q^\top]^\top$, each power-flow equation is a quadratic $x^\top C_i x = b_i$. These native surfaces are unbounded (hyperbolic), so raw branch tracing can run off to infinity. Following Lesieutre–Wu (2015) and Wu’s thesis, a positive-definite *base ellipsoid* is added to every equation, producing an equivalent system

$$x^\top M_i x = 1, \quad M_i = M_i^\top \succ 0, \quad i = 1, \dots, 2N_{\text{bus}},$$

whose zero set is *identical* to the original (algebraic-set preserving) but whose every equation is now a high-dimensional ellipse centred at the origin. Relaxing one equation with a tracing parameter α (the code variable `aal`) yields a 1-dimensional curve; because the ellipses are

compact, each connected component of that curve is a **bounded loop** (generically diffeomorphic to a circle), so a trace must return to its start. Solutions occur wherever $\alpha = 0$ along the loop.

6.2 Holomorphic embedding and Padé

Rather than crawl the loop with small predictor–corrector steps, HEBC embeds the tracing parameter into the complex plane and represents each bus voltage as a holomorphic function, computed as a power series (to degree `degree.act`, default 15). All series degrees share a single LU factorisation, so the coefficients are cheap. A **Padé approximant** (balanced numerator/-denominator degree, for maximal convergence radius) then extrapolates the voltages far along the curve in one step, until the power mismatch exceeds `param.mismatch_error`. The real roots of the Padé denominator reveal the location of the nearest singularity on the curve.

6.3 Hybrid switching and warm start

When a holomorphic step approaches a singularity — signalled either by the corrector failing to converge, or by the holomorphic increment $|\Delta\alpha|$ collapsing — the solver switches to predictor–corrector continuation to pass through the fold. To avoid a slow “cold” warm-up, a **warm starter** sets the first continuation step to a fraction of the estimated distance to the singularity (and no larger than a fraction of the last holomorphic step), then builds a quadratic predictor. After the fold is cleared, control returns to the holomorphic routine. One holomorphic step costs about $4\times$ a continuation step but typically covers ≈ 8 – 9 continuation steps of arc, which is where the speed-up comes from.

6.4 Continuation details

The continuation phase uses a pseudo-arclength predictor (quadratic, from the last three points), a Newton corrector (*Phase I*), and a variable-rescaling corrector (*Phase II*) for badly-conditioned near-singular arcs whose secant slope exceeds `param.slope_max`. The step length adapts to the corrector iteration count toward a target of `param.targetsteps`. Each α sign change triggers a solution extraction and a duplicate check; a trace terminates when it returns to its starting point (within `closure` tolerance) or exhausts the alternation cap `param.outnum`.

7 Parameter reference

Parameters live in the `param` struct built at the top of `hybrid_traceloops_4_with_mex.m` (and mirrored in the parallel drivers). Defaults below are the shipped values of this legacy-fast + consensus build.

7.1 Continuation / step control

Parameter	Default	Meaning
<code>sstep</code>	1e-5	Base arc-length step scale for the continuation.
<code>targetsteps</code>	3	Target number of corrector iterations; drives step-size adaptation.
<code>imax</code>	40	Maximum Newton iterations per corrector step.
<code>smax</code>	20000	Maximum continuation steps per <code>branch_trace</code> call.
<code>maximumstep</code>	8000	Maximum step-size multiplier (Phase I ceiling base).
<code>maxphase1</code>	<code>maximumstep/2</code>	Maximum step in Phase I.

Parameter	Default	Meaning
maxphase2	maxphase1/10	Maximum step in Phase II.
minimumstep	1e-2	Minimum step-size multiplier.
ef	1e-8	Corrector residual tolerance $\ F\ $.
ex	1e-7	Corrector step tolerance $\ \Delta x\ $.
PS	300	Phase II (rescaling) duration, in steps.
maxsolu	100	Maximum solutions collected per traced loop.
maxskip	50	Consecutive numerical stalls tolerated before the loop is skipped.
dev	20	Console print interval (display only).

7.2 Holomorphic / hand-off control

Parameter	Default	Meaning
mismatch_error	1e-4	Maximum holomorphic power-mismatch accepted before shrinking the arc.
holonum	50	Maximum Padé steps per outer iteration before forcing continuation.
holo_arc_max	0.01	Maximum holomorphic arc length per step.
outnum	500	Cap on holomorphic $\leftrightarrow$ continuation alternations per curve (a stall burns this cap).
slope_max	4e5	Maximum secant slope at which a turning point is accepted as passed. Lower (e.g. 1e4) is more robust but slower.
aal_min	1e-6	Minimum homotopy-parameter increment; the unit for handback .
im_tol	1e-3	Imaginary-part tolerance for accepting a Padé root as a real pole.
handback	0.1	Hand-back hysteresis ($\times$ aal_min): how far past a turning point to travel before returning to the holomorphic routine. 0.1 suits case39/case30; 0.5 suits case3/case118.
interval	1	Stride for the turning-point finite-difference test.

7.3 Consensus pole estimation

Parameter	Default	Meaning
pole_mode	'random'	'fixed' = predefined bus set; 'random' = random set drawn once per curve.
pole_fixed_buses	1:2	Bus set used when pole_mode='fixed' (auto-filtered to $\leq$ bus count).
pole_nbus	3	Number of buses drawn when pole_mode='random'.
pole_consensus	true	If true, take the smallest pole agreed on by $\geq$ half the sampled buses; else global minimum.
pole_cl_tol	0.10	Relative magnitude tolerance for clustering "the same pole" across buses.

7.4 Holomorphic degree

Set in the drivers / `main.m`: `degree.act = 15` (active power-series degree), split as `degree.numerator = 8`, `degree.denominator = 7` for the Padé approximant. **The shipped MEX bake these degree values in as compile-time constants** — if you change them, regenerate `examples.mat` and rebuild the MEX (Section 10).

8 Outputs

Console: per-case progress, solution count, and CPU (serial) or wall (parallel) time, followed by a summary table.

CSV summaries (written into the run folder):

Driver	CSV file
<code>run_batch.m</code>	<code>results_summary.csv</code>
<code>run_batch_par.m</code>	<code>results_par_summary.csv</code>
<code>run_batch_parfeval.m</code>	<code>results_parfeval_summary.csv</code>

Columns: `case_name`, `cpu_time_sec` (or wall time), `n_solutions`, `status` (OK / ERROR), `error_msg`. These CSV files are run outputs and are ignored by the repository `.gitignore`.

The full solution set is available in the workspace as `Zsave` (each column a solution in the internal variable ordering) after a run; the solver sign-normalises and deterministically orders it (`canonicalize_solutions.m`) so that serial, parfor and parfeval produce identical output for the same case. Antipodal pairs $\{x, -x\}$ are identified as one solution.

9 File map

Setup / preprocessing: `makeYbus` from MATPOWER (Ybus), `get_quadr_mtrx.m` (power-balance quadratics), `quadr_matrix.m` (per-bus PV/PQ/slack matrices), `Solu.m` (initial point from `runpf`), `MakeEllipse.m` (base-ellipse generation), `Tune_Fac.m` (continuation scaling), `main.m` (reference preprocessing script; its logic is inlined into the drivers).

Core solver (shared hot path): `hybrid_traceloops_4_with_mex.m` (outer trace-loop orchestrator; builds `param`, seeds the RNG, primes the cached operators), `holomorphic_hybrid_5_with_mex.m` (one holomorphic step + consensus pole estimate), `branch_trace_hybrid_4_with_mex.m` (predictor-corrector continuation), `Resolve_with_mex.m` (Newton corrector), `holomorphic_cont_tri.m(+MEX)` (Taylor-coefficient triangular solves), `holomorphic_para_sp.m` (sparse H builder), `Pade_Apprxmt.m(+MEX)` (Padé construction), `update_V_Pade.m` (Padé evaluation along the arc), `MakeJacobianD.m` (cached sparse Jacobian), `quad_form_vals.m` (cached constraint-residual evaluator), `MakeJacobian.m` (scalar Jacobian, used by `Tune_Fac`).

Post-processing: `canonicalize_solutions.m` (sign-normalise + deterministic ordering).

Parallel housekeeping: `cleanup_parallel_artifacts.m` (clears per-worker caches, optionally closes the pool to avoid a Windows/R2022a `-batch` shutdown crash).

Drivers: serial — `runVBook_hybrid_2023.m`, `run_batch.m`; parfor — `runVBook_hybrid_parallel.m`, `trace_equation_worker.m`, `run_batch_par.m`; parfeval — `runVBook_hybrid_parfeval.m`, `trace_equation_worker_const.m`, `trace_equation_worker.m`, `run_batch_parfeval.m`; convenience — `run_merged_case.m` (one case, any mode).

Build: `build_mex.m`, `examples.mat` (codegen type seeds).

Case data: caseXXX.m MATPOWER-format test systems.

10 Rebuilding the MEX

The shipped binaries are Windows 64-bit. On other platforms, or after editing any hot-path source, rebuild with MATLAB Coder:

```
>> cd '<path>\HEBCPF_MEX_v2.202607'
>> addpath(pwd)
>> build_mex      % rebuilds both .mexw64 from the .m source + examples.mat
```

build_mex.m defines the codegen argument types (degree as coder.Constant; the I.pv/I.pq index fields typed variable-size because case3 has no PV bus). If you change degree.act/numerator/denominator regenerate examples.mat (save the degree/I/Start structs from any quick run) and rebuild, because the MEX are specialised to those constant degree values.

Windows build gotcha. If codegen fails with “Pade_Apprxmt_mex.bat is not recognized”, the environment variable NoDefaultCurrentDirectoryInExePath is set, blocking codegen’s generated .bat. Unset it *before* launching MATLAB:

```
bash :  unset NoDefaultCurrentDirectoryInExePath ; matlab ...
cmd   :  set "NoDefaultCurrentDirectoryInExePath=" && matlab ...
```

Also try mex -setup C and re-selecting the compiler (developed with MSVC 2019). If you have no compiler, use the pure-MATLAB build — same answers, $\sim 2\times$ slower.

To run *without* the MEX (A/B comparison), comment the _mex call lines in holomorphic_hybrid_5_with_mex.m and uncomment the corresponding .m calls.

11 Troubleshooting

Undefined function 'runpf'

MATPOWER is not on the MATLAB path. Install and addpath it.

Undefined function 'makeYbus'

MATPOWER is not on the MATLAB path, or the MATPOWER lib folder is missing from the path.

Undefined function 'caseXXX'

Run addpath(pwd) from this folder; the case data files live here.

Undefined function 'Pade_Apprxmt_mex' / “generated for a different platform”

You are not on Windows x64, or the binary moved. Run build_mex, or use a pure-MATLAB build.

codegen ".bat is not recognized"

See the Windows build gotcha (Section 10).

OSQP warning during runpf

Harmless MATPOWER solver-probe message; runpf still converges (equation error ~ 0).

Parallel driver appears to hang at first run

The local parpool is starting; this one-time cost is excluded from the reported timing.

Solution count differs from the expected value

Check the load factor is 1 and that the MATPOWER initial solve (`runpf`) reported success. Note that the `pole_mode/handback` settings affect *tracing efficiency and numerical robustness*, not the final solution set on well-behaved cases.

12 License and third-party notices

HEBCPF is distributed under the BSD 3-Clause License; see the top-level LICENSE file. Some bundled `caseXXX.m` files are copied from, derived from, or modified from MATPOWER and other public test-system sources. Preserve the per-file attribution notices when redistributing modified versions, and see the top-level NOTICE file for third-party attribution details.

13 Citing this solver

If you use results produced by this solver, please cite the HEBC paper:

D. Wu and B. Wang, “Holomorphic Embedding Based Continuation Method for Identifying Multiple Power Flow Solutions,” *IEEE Access*, vol. 7, pp. 86843–86853, 2019. doi:[10.1109/ACCESS.2019.2925384](https://doi.org/10.1109/ACCESS.2019.2925384).

BibTeX:

```
@article{wu2019hebc,
  author = {Wu, Dan and Wang, Bin},
  title = {Holomorphic Embedding Based Continuation Method for
    Identifying Multiple Power Flow Solutions},
  journal = {IEEE Access},
  volume = {7},
  pages = {86843--86853},
  year = {2019},
  doi = {10.1109/ACCESS.2019.2925384}
}
```

For citing the software package itself, see the top-level CITATION.cff file.

Further reading

The method rests on, and is extended by, the following works:

1. B. C. Lesieutre and D. Wu, “An Efficient Method to Locate All the Load Flow Solutions — Revisited,” in *2015 53rd Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, Monticello, IL, USA, pp. 381–388, IEEE, 2015. doi:[10.1109/ALLERTON.2015.7447029](https://doi.org/10.1109/ALLERTON.2015.7447029). (*The elliptical branch-tracing formulation that underlies this solver.*)
2. D. Wu, “Algebraic Set Preserving Mappings for Electric Power Grid Models and Its Applications,” Ph.D. dissertation, University of Wisconsin–Madison, 2017. (*The dissertation that formalises the elliptical / algebraic-set-preserving formulation and specifies the predictor–corrector continuation machinery — the Phase I/II correctors, step-length control, and solution/termination logic — that this solver implements.*)

3. D. Wu, D. K. Molzahn, B. C. Lesieutre, and K. Dvijotham, “A Deterministic Method to Identify Multiple Local Extrema for the AC Optimal Power Flow Problem,” *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 654–668, Jan. 2018. doi: [10.1109/TPWRS.2017.2707925](https://doi.org/10.1109/TPWRS.2017.2707925). (*Extends the tracing method to AC optimal power flow.*)
4. B. Wang, D. Wu, X. Su, K. Sun, and L. Xie, “A Long-Term Voltage Stability Margin Index Based on Multiple Real Power Flow Solutions,” in *2023 IEEE Power & Energy Society General Meeting (PESGM)*, Orlando, FL, USA, pp. 1–5, IEEE, Jul. 2023. doi: [10.1109/PESGM52003.2023.10252776](https://doi.org/10.1109/PESGM52003.2023.10252776). (*Defines a voltage-stability margin as the minimum voltage-space distance from the high-voltage solution to all other real power-flow solutions — a direct application of this solver’s output.*)
5. D. Wu and B. Wang, “Influence of Load Models on Equilibria, Stability and Algebraic Manifolds of Power System Differential-Algebraic System,” in *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, Monticello, IL, USA, IEEE, 2019. doi: [10.1109/ALLERTON.2019.8919649](https://doi.org/10.1109/ALLERTON.2019.8919649). (*How load models reshape the solution manifold and its equilibria.*)
6. D. Wu and B. Wang, “Collection of Numerous Power Flow Solutions of Standard IEEE Test Systems,” *IEEE Dataport*, Jul. 2019. doi: [10.21227/24bh-hj72](https://doi.org/10.21227/24bh-hj72). (*The published dataset of reference solution sets for the standard IEEE test systems — useful for validating this solver’s output.*)